

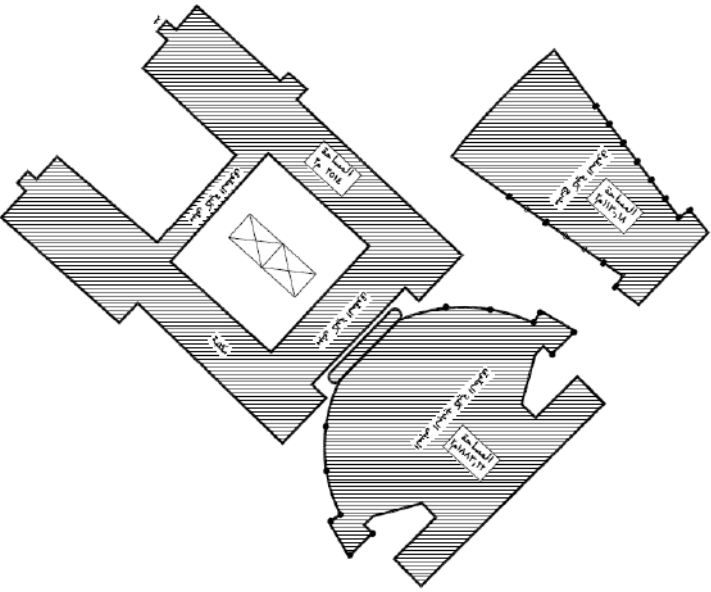
PV On-Grid System

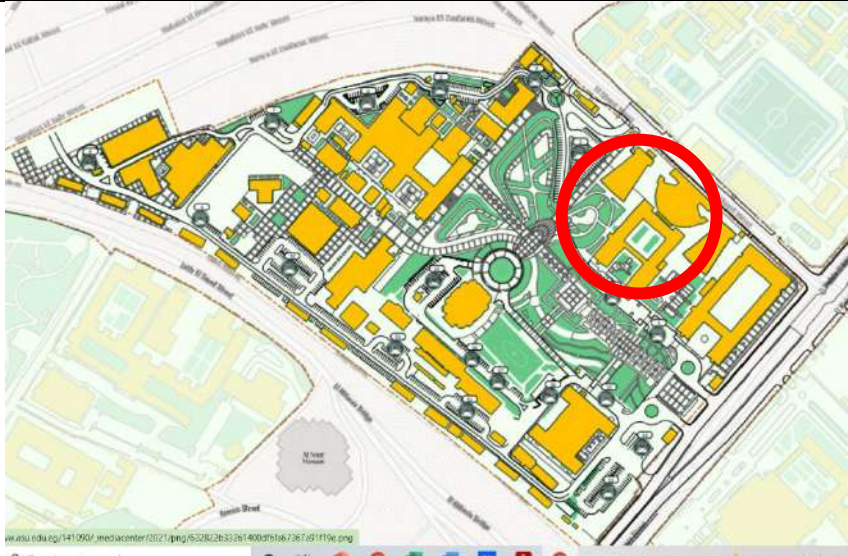
Report Contents:

1. University Building Identification.
2. Radiation Data.
3. RETScreen Report.
4. PV Module and Inverter Selection and Characteristics.
5. System Sizing and Design Calculations.
6. PV System Summary.
7. PVsyst Report.

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University Building Identification.

University Name:	Ain Shams University
Location (City Name and GPS Coordinates)	Cairo.
Title of Building* ¹ where the EE/RE opportunities will be implemented:	Faculty of Law.
Building Address:	Campus (1) - Ain Shams University - El-Khalifa El-Mamoun El-Abbasyia, Cairo, Egypt
Estimated Annual Building Energy Consumption (MWh): Contribution of Building's Energy Consumption to Overall Energy Consumption of the Faculty and University: (ratio x:y)	Building Energy Consumption / Faculty Energy Consumption = 629 MWh/Year Approximate Faculty/ University Campus 1 Consumption % = 12 %
Building Location:	 A2. Faculty of Law.

	 Faculty of Law - Circled in RED
Building Footprint Area (m²)	Main Building == > 4397 m2 & Lecture Hall ==> 1,130.18 m2
Building Number of Floors	Basement & Four Floors
Building Functions, i.e. Admin, Faculty, Department, Lectures, Lab, Library, Workshop, Student Services,..... etc.	Admin, Faculty, Department, Lectures, Lab, Library, Workshop

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Radiation data

Based on PVGIS Tool [Photovoltaic Geographical Information System]

Fixed mounting System

Cursor: 30.095, 31.314
Selected: 30.077, 31.286
Elevation (m): 29
PVGIS ver. 5.2

Use terrain shadows:
☒ Calculated horizon
☐ Upload horizon file
[Switch to version 5.1](#)

[Download CSV](#)
Choose File

[Download JSON](#)
No file chosen

GRID CONNECTED

TRACKING PV

OFF-GRID

MONTHLY DATA

DAILY DATA

HOURLY DATA

TMY

 **PERFORMANCE OF GRID-CONNECTED PV** 

Solar radiation database*
PV technology*
Installed peak PV power [kWp]*
System loss [%]*

PVGIS-SARAH2

Crystalline silicon

207

15

Fixed mounting options

Mounting position *
Slope [°]*
Azimuth [°]*

Free-standing

35

0

☐ **PV electricity price**
PV system cost (your currency)
Interest [%/year]
Lifetime [years]

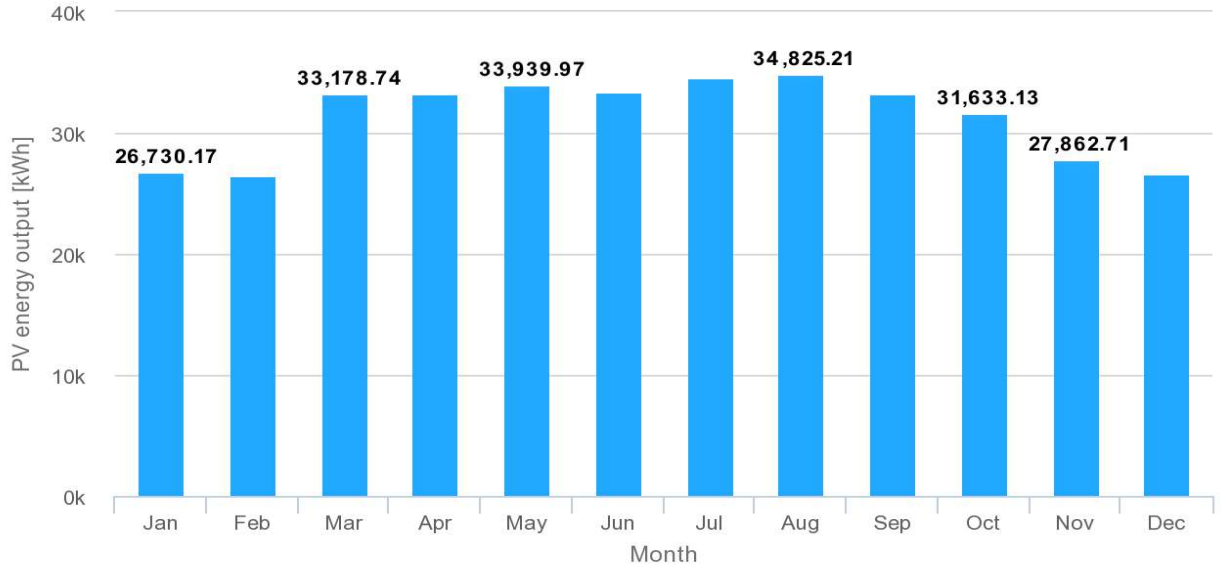
☐ Optimize slope
☒ Optimize slope and azimuth

Summary

Provided inputs:	
Location [Lat/Lon]:	30.077,31.286
Horizon:	Calculated
Database used:	PVGIS-SARAH2
PV technology:	Crystalline silicon
PV installed [kWp]:	207
System loss [%]:	15
Simulation outputs:	
Slope angle [°]:	28 (opt)
Azimuth angle [°]:	5 (opt)
Yearly PV energy production [kWh]:	375910.27
Yearly in-plane irradiation [kWh/m ²]:	2429.15
Year-to-year variability [kWh]:	3446.61
Changes in output due to:	
Angle of incidence [%]:	-2.48
Spectral effects [%]:	0.05
Temperature and low irradiance [%]:	-9.85
Total loss [%]:	-25.24

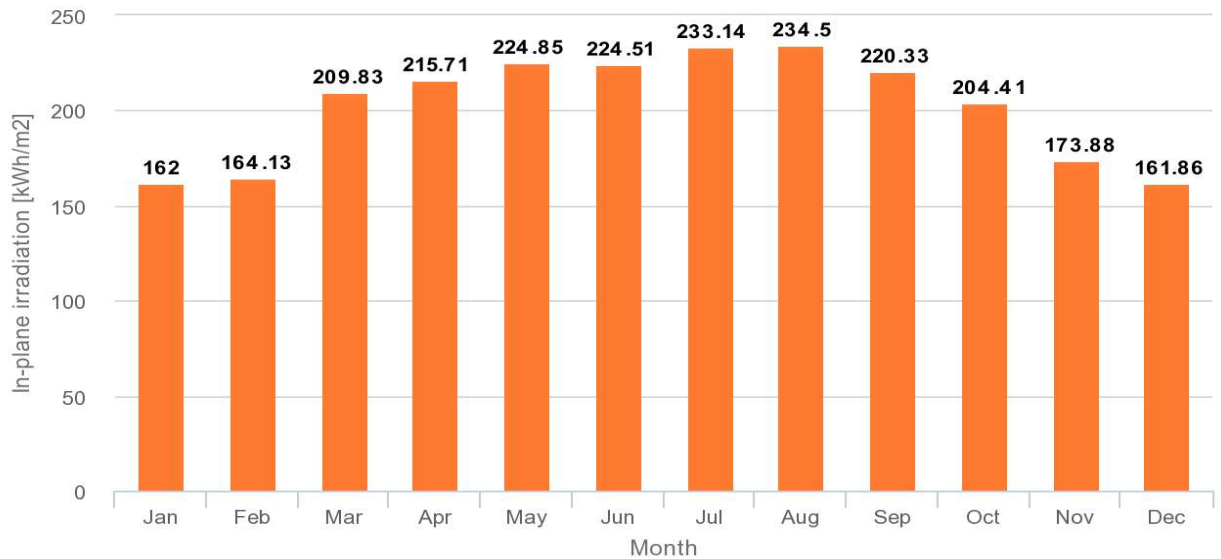
Monthly energy output from fix-angle PV system

(C) PVGIS, 2023



Monthly in-plane irradiation for fixed angle

(C) PVGIS, 2023



Mounting	Yearly PV energy production [kWh]
Fixed	375910.27

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RETScreen Report

Feasibility report

Ain Shams University

200 kW



Power plant - Photovoltaic

Prepared for:

Graduation Project

Prepared by:

Mohamed Tawheed

Executive summary

This report was prepared using the RETScreen Clean Energy Management Software. The key findings of this analysis are presented

Target

	Electricity exported to grid MWh	Electricity export revenue \$	GHG emission reduction tCO ₂
Proposed case	378	37,760	183

Site reference conditions

Climate data location
Egypt - Cairo H.Q.

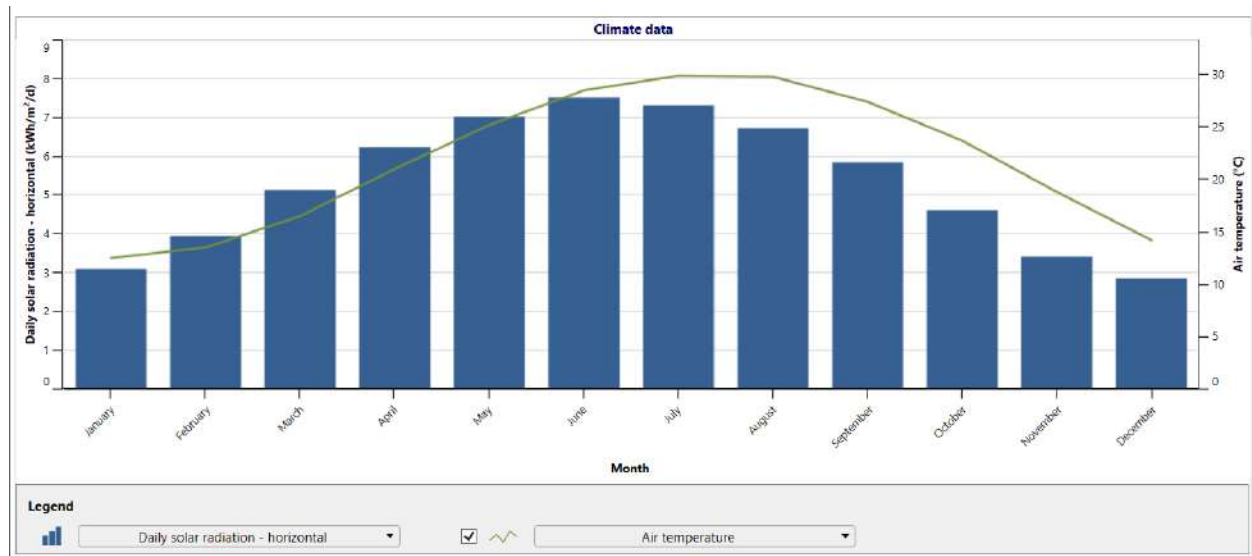
Facility location
Egypt - Cairo - Qesm El Waily

Legend
 Facility location
 Climate data location



	Unit	Climate data location	Facility location	Source
Latitude		30.1	30.1	
Longitude		31.3	31.3	
Climate zone		2B - Hot - Dry		
Elevation	m	26	28	NASA
Heating design temperature	°C	7.0		Ground - Map
Cooling design temperature	°C	34.7		NASA
Earth temperature amplitude	°C	21.4		NASA

Month	Air temperature °C	Relative humidity %	Precipitation mm	Daily solar radiation - horizontal kWh/m ² /d	Atmospheric pressure kPa	Wind speed m/s	Earth temperature °C	Heating degree-days 18 °C °C-d	Cooling degree-days 10 °C °C-d
January	12.5	60.5%	8.37	3.09	100.9	3.6	12.5	171	78
February	13.5	54.4%	7.00	3.94	100.8	3.8	13.7	126	98
March	16.5	48.1%	7.44	5.13	100.6	4.0	17.4	47	202
April	21.0	39.0%	3.00	6.23	100.4	4.2	22.5	0	330
May	25.2	35.2%	0.93	7.02	100.3	4.3	27.4	0	471
June	28.5	34.8%	0.00	7.51	100.1	4.4	31.0	0	555
July	29.9	37.0%	0.00	7.31	99.8	4.0	32.4	0	617
August	29.8	39.9%	0.00	6.72	99.9	3.7	32.1	0	614
September	27.4	45.0%	0.30	5.84	100.2	3.9	29.2	0	522
October	23.7	51.0%	1.86	4.61	100.5	3.7	25.0	0	425
November	18.8	56.0%	4.50	3.41	100.7	3.6	19.3	0	264
December	14.2	60.3%	6.20	2.85	100.9	3.6	14.1	118	130
Annual	21.8	46.7%	39.60	5.31	100.4	3.9	23.1	461	4,305
Source	NASA	NASA	NASA	Ground	NASA	NASA	NASA	NASA	NASA
Measured at					m	10	0		



Power plant

Photovoltaic
200 kW



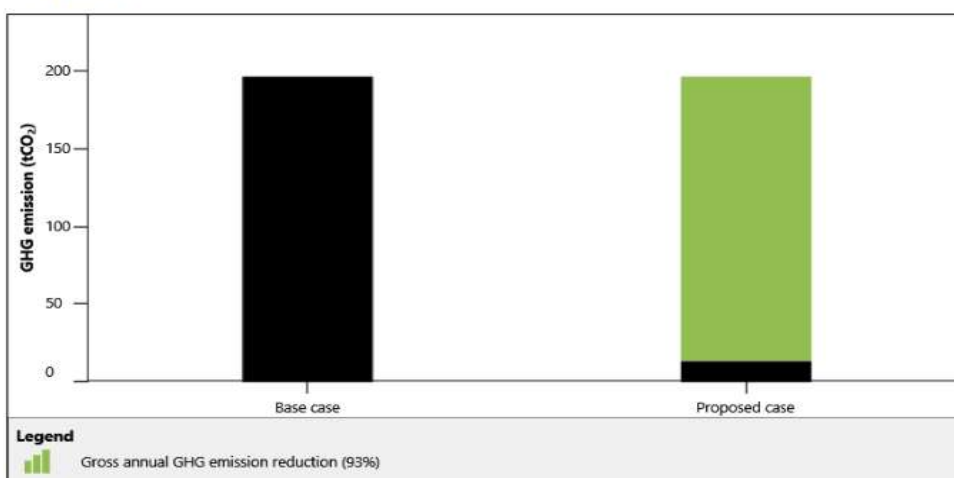
Photovoltaic - 200 kW		
Capacity	207	kW
Electricity	378	MWh

Target

Summary

	Electricity exported to grid MWh	Electricity export revenue \$	GHG emission reduction tCO ₂
Proposed case	378	37,760	183

GHG emission



GHG equivalence



183.1 tCO₂ is equivalent to 78,689
Litres of gasoline not consumed

GHG emission

Base case	196.9	tCO ₂
Proposed case	13.8	tCO ₂
Gross annual GHG emission reduction	183.1	tCO₂

Financial viability

Financial parameters

General

Inflation rate	%	2%
Discount rate	%	9%
Reinvestment rate	%	9%
Project life	yr	25

Finance

Debt ratio	%	70%
Debt	\$	144,900
Equity	\$	62,100
Debt interest rate	%	7%
Debt term	yr	15
Debt payments	\$/yr	15,909

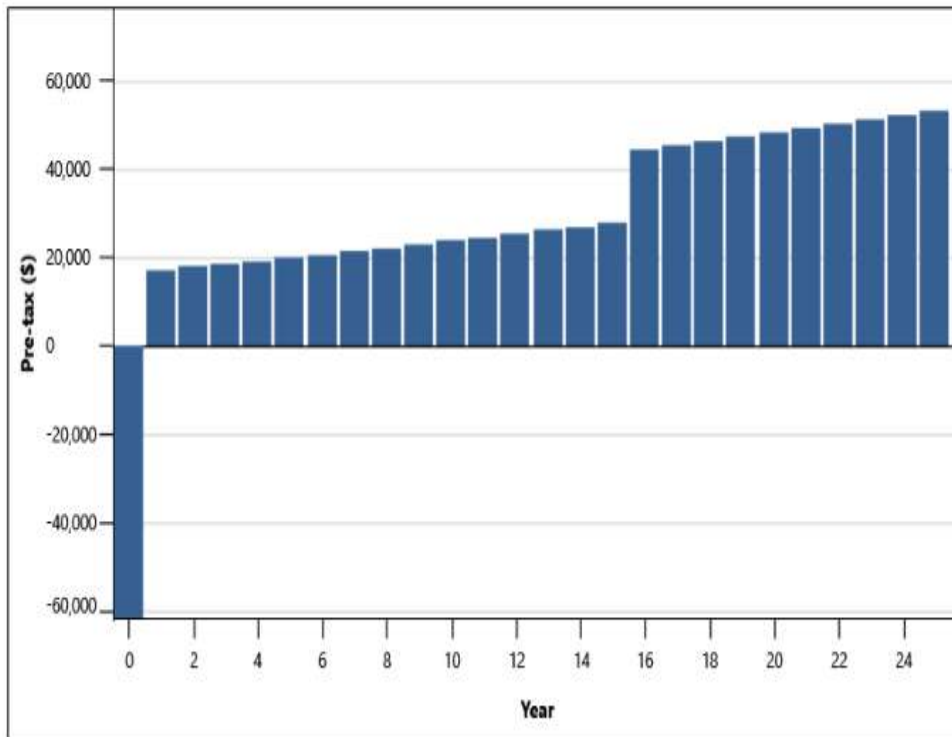
Annual revenue

Electricity export revenue

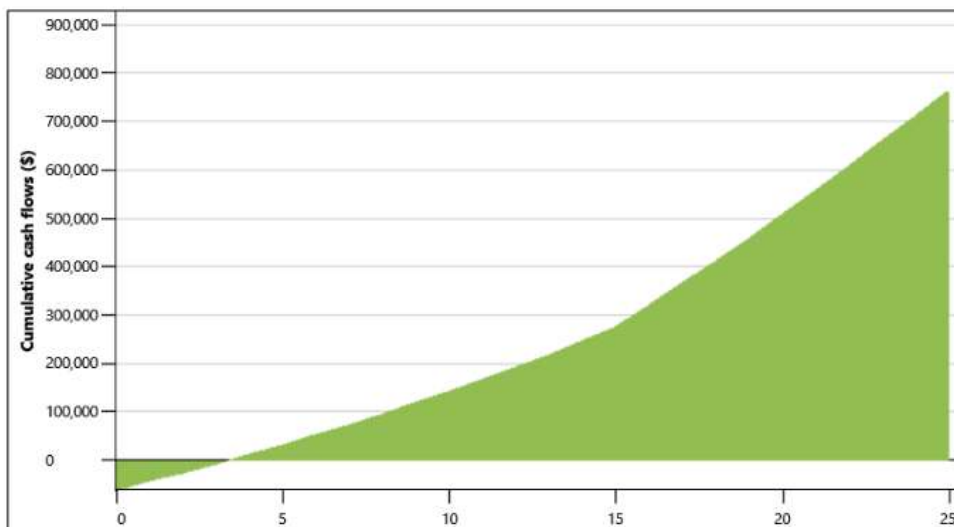
Electricity exported to grid	MWh	378
Electricity export rate	\$/kWh	0.10
Electricity export revenue	\$	37,760
Electricity export escalation rate	%	2%

Cash flow

Annual



Cumulative



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PV SUNTECH module Specifications [Data Sheet]

Electrical Characteristics

STC	STPXXXS-C72/Vmh				
Maximum Power at STC (Pmax)	550W	545W	540W	535W	530W
Optimum Operating Voltage (Vmp)	42.05V	41.87V	41.75V	41.57V	41.39V
Optimum Operating Current (Imp)	13.08A	13.02A	12.94A	12.87A	12.81A
Open Circuit Voltage (Voc)	49.88V	49.69V	49.54V	49.39V	49.24V
Short Circuit Current (Isc)	14.01A	13.96A	13.89A	13.83A	13.76A
Module Efficiency	21.3%	21.1%	20.9%	20.7%	20.5%
Operating Module Temperature	-40 °C to +85 °C				
Maximum System Voltage	1500 V DC (IEC)				
Maximum Series Fuse Rating	25 A				
Power Tolerance	0/+5 W				

STC: Irradiance 1000 W/m², module temperature 25 °C, AM=1.5;
Tolerance of Pmax is within +/- 3% ;
For tracker installation, please turn to Suntech for mechanical load information.

NMOT	STPXXXS-C72/Vmh				
Maximum Power at NMOT (Pmax)	415.0W	411.5W	408.0W	404.3W	400.6W
Optimum Operating Voltage (Vmp)	38.9V	38.7V	38.6V	38.4V	38.2V
Optimum Operating Current (Imp)	10.67A	10.63A	10.58A	10.53A	10.47A
Open Circuit Voltage (Voc)	46.9V	46.7V	46.5V	46.4V	46.3V
Short Circuit Current (Isc)	11.22A	11.18A	11.13A	11.08A	11.02A

NMOT: Irradiance 800 W/m², ambient temperature 20 °C, AM=1.5, wind speed 1 m/s.

Temperature Characteristics

Nominal Module Operating Temperature (NMOT)	42 ± 2 °C
Temperature Coefficient of Pmax	-0.36%/°C
Temperature Coefficient of Voc	-0.304%/°C
Temperature Coefficient of Isc	0.050%/°C

Mechanical Characteristics

Solar Cell	Monocrystalline silicon 182 mm
No. of Cells	144 (6 × 24)
Dimensions	2279 × 1134 × 35 mm (89.7 × 44.6 × 1.4 inches)
Weight	29.1 kgs (64.2 lbs.)
Front Glass	3.2 mm (0.126 inches)
Frame	Anodized aluminium alloy
Junction Box	IP68 rated (3 bypass diodes)
Output Cables	4.0 mm ² , Portrait: (-) 350 mm and (+) 160 mm in length Landscape : (-) 1 400 mm and (+) 1400 mm in length or customized length
Connectors	MC4 EVO2, Cable 01S

SUNTECH Module Characteristics

Maximum Power at STC ($P_{\max}$)= 540W.

Current at $P_{\max}$ = 12.94 A

Voltage at $P_{\max}$ = 41.75 V

S.C Current = 13.89 A

O.C Voltage = 49.54 V

Maximum System Voltage = 1500 V DC (IEC)

Nominal Module Operating Temperature (NMOT) = 42 ± 2 °C

Dimensions = 2279 mm* 1134 mm* 35 mm

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Huawei Smart String Inverter Specifications [Data Sheet]



Smart String Inverter (SUN2000-60KTL-HV-D1-001)

Technical Specifications	SUN2000-60KTL-HV-D1-001
	Efficiency
Max. Efficiency	99.0%
European Efficiency	98.8%
	Input
Max. Input Voltage	1,500 V
Max. Current per MPPT	22 A
Max. Short Circuit Current per MPPT	30 A
Start Voltage	650 V
MPPT Operating Voltage Range	600 V ~ 1,450 V
Rated Input Voltage	1,080 V
Number of Inputs	8
Number of MPP Trackers	4
	Output
Rated AC Active Power	60,000 W
Max. AC Apparent Power	66,000 VA
Max. AC Active Power (cosφ=1)	66,000 W
Rated Output Voltage	800 V, 3W + PE
Rated AC Grid Frequency	50 Hz / 60 Hz
Rated Output Current	43.3 A
Max. Output Current	48 A

Huawei – SUN2000-60KTL-HV-D1-001 Inverter Characteristics

Unit Nominal Power = 60 KW AC

DC Voltage Range MPPT = 600 V – 1450 V

Max. Input Voltage = 1500 V

Max. Current per MPPT = 22 A

Max. Short Circuit Current per MPPT = 30 A

Number of Inputs = 8

Number of MPP Trackers = 4

PF = 0.9

$\eta_{Inv} = 99 \%$

Panel Protection: IP65

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Sizing and Design of Solar PV System

$$\text{Number of modules} = \frac{\text{Solar PV system power (KW)} * 1,000}{\text{PV module power (W)}}$$

$$\text{Number of modules} = \frac{207 * 1,000}{540} = \mathbf{384 \text{ Module}}$$

For ground installation with row spacing, 1 kW of Solar PV system needs approximately: 8-10 m²

$$\text{PV System Area} = 207 * [8 - 10] = [\mathbf{1656 - 2070}] \text{ m}^2$$

Number of modules per string

From max and min ambient temperatures, get max and min cell temperatures:

$$T_{\text{cell max}} (^{\circ}\text{C}) = T_{\text{a max}} + 25$$

$$T_{\text{cell max}} (^{\circ}\text{C}) = 45 + 25 = \mathbf{70^{\circ}\text{C}}$$

$$T_{\text{cell min}} (^{\circ}\text{C}) = T_{\text{a min}} + 25$$

$$T_{\text{cell min}} (^{\circ}\text{C}) = 5 + 25 = \mathbf{30^{\circ}\text{C}}$$

Correction of PV module MPP voltage to get $V_{mpp \max}$ and $V_{mpp \min}$

For the PV module:

$$V_{mpp \min} = V_{mpp} + [(T_{cell \max} - 25) * T_c * V_{oc}] \quad \text{When } T_c (\%/^{\circ}C)$$

$$V_{mpp \min} = 41.75 + [(70 - 25) * -0.304\%/^{\circ}C * 49.54] = \mathbf{35 \text{ V}}$$

$$V_{mpp \max} = V_{mpp} + [(T_{cell \min} - 25) * T_c * V_{oc}] \quad \text{When } T_c (\%/^{\circ}C)$$

$$V_{mpp \max} = 41.75 + [(30 - 25) * -0.304\%/^{\circ}C * 49.54] = \mathbf{40.95 \text{ V}}$$

$$\frac{\text{Inverter } V_{mpp \min}}{\text{PV module } V_{mpp \min}} < \text{No. of Modules per String} < \frac{\text{Inverter } V_{mpp \max}}{\text{PV module } V_{mpp \max}}$$

$$\frac{600 \text{ V}}{35 \text{ V}} = 17.14 < \text{No. of Modules per String} < 35.41 = \frac{1450 \text{ V}}{40.95 \text{ V}}$$

$$\mathbf{17 \leq \text{No. of Modules per String} \leq 35}$$

Correction of PV module open circuit voltage to get $V_{OC\ max}$

For the PV module:

$$V_{oc\ max} = V_{oc} + [(T_{cell\ min} - 25) * T_c * V_{oc}] \quad \text{When } T_c (\%/^{\circ}C)$$

$$V_{oc\ max} = 49.54 + [(30 - 25) * -0.304\%/^{\circ}C * 49.54] = \mathbf{48.75\ V}$$

$$\text{No. of Modules per String} < \frac{\text{Inverter } V\ max}{\text{PV module } V_{oc\ max}}$$

$$\text{No. of Modules per String} < 29.13 = \frac{1420\ V}{48.75\ V}$$

$$\text{No. of Modules per String} \leq \mathbf{29}$$

$$\text{No. of Modules per String} = \mathbf{24}$$

Number of strings

Correction of PV module short circuit current to get $I_{sc\ max}$.

For the PV module:

$$I_{sc\ max} = I_{sc} + [(T_{cell\ max} - 25) * T_c * I_{sc}] \quad \text{When } T_c (\%/^{\circ}C)$$

$$I_{sc\ max} = 13.89 + [(70 - 25) * 0.050\%/^{\circ}C * 13.89] = \mathbf{14.2\ Amp}$$

$$\text{No. of Strings per MPPT} < \frac{\text{Inverter } I\ max}{\text{PV module } I_{sc\ max}}$$

$$\text{No. of Strings per MPPT} < 1.55 = \frac{22\ Amp}{14.2\ Amp}$$

$$\text{No. of Strings per MPPT} \leq \mathbf{1}$$

$$\text{Inverter total strings} = 1 \frac{\text{string}}{\text{MPPT}} * \text{no. of MPPTs} = 1 * 4 = \mathbf{4\ strings}$$

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PV System Summary

PV System Power = **207 kW**

No. of Modules = 207 kW / 540 W = **384** Module

1. No. of Modules in series = **24** Module

2. No. of strings in parallel = **16** String

System Voltage = 24 * 40.95 = **983 V**

System Current = 4 * 12.94 = **52 Amp**

No. of Inverters = 207 kW / 60 kW = **4** Inverters

Total Power of Inverters = **240 kW AC**

$\eta_{PV} = 20.9 \%$

$\eta_{Inv} = 99 \%$

$\eta_{Cables} = 98 \%$

Utilized Area = 2514 **m²**

Available area percentage = 80 %

PV System Area = 2011 **m²**

kW of PV modules cost = **\$1000 / kW**

PV system cost = **\$207,000**

Annual Energy Production = **341,100 kWh**

Annual Saving = Energy * Tariff = 341,100 * 1.15 (kWh/LE) = **392,265 LE**

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PVsyst Report

Grid system presizing

Geographical Site	Cairo	Country	Egypt
Situation	Latitude 30.08° N	Longitude	31.29° E
Time defined as	Legal Time Time zone UT+2	Altitude	0 m
Collector Plane Orientation	Tilt 28°	Azimuth	5°

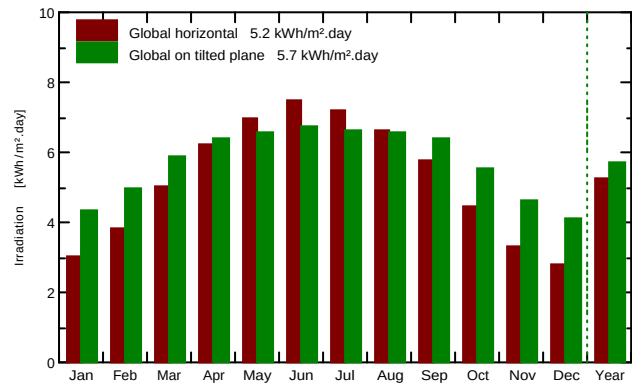
PV-field installation main features

Module type	Standard
Technology	Monocrystalline cells
Mounting method	Flat roof
Back ventilation properties	Free standing

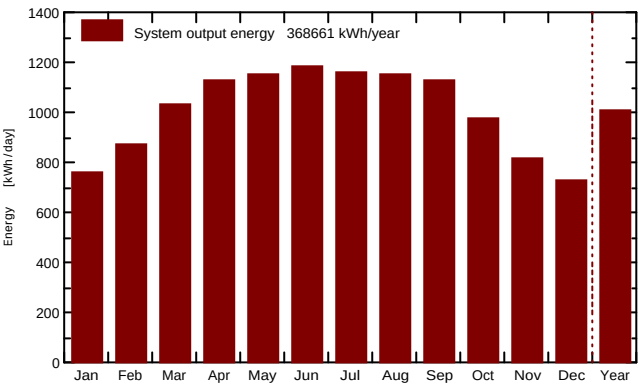
System characteristics and pre-sizing evaluation

PV-field nominal power (STC)	Pnom	207 kWp		
Collector area	Acoll	1294 m²		
Annual energy yield	Eyear	369 MWh	Specific yield	1781 kWh/kWp
Economic gross evaluation	Investment	468306 EUR	Energy price	0.08 EUR/kWh

Meteo and incident energy



System output



	Gl. horiz. kWh/m².day	Coll. Plane kWh/m².day	System output kWh/day	System output kWh
Jan.	3.02	4.32	760.1	23564
Feb.	3.84	4.97	874.5	24487
Mar.	5.06	5.88	1034	32040
Apr.	6.22	6.43	1131	33919
May	6.95	6.57	1155	35795
June	7.49	6.75	1187	35598
July	7.23	6.63	1165	36120
Aug.	6.61	6.56	1154	35767
Sep.	5.76	6.41	1127	33814
Oct.	4.48	5.58	980.4	30393
Nov.	3.34	4.66	819.0	24569
Dec.	2.78	4.15	728.9	22595
Year	5.24	5.75	1010	368661

PVSYST V6.81					04/05/23	Page 1/5
Grid-Connected System: Simulation parameters						
Project : Tawheed GP 1						
Geographical Site			Faculty of Law, Ain Shams University		Country	Egypt
Situation		Latitude	30.08° N	Longitude	31.29° E	
Time defined as		Legal Time	Time zone UT+2		Altitude	0 m
		Albedo	0.20			
Meteo data:		Faculty of Law, Ain Shams University		Meteonorm 7.2 (1981-2009) - Synthetic		
Simulation variant : New simulation variant						
			Simulation date	04/05/23 21h49		
Simulation parameters		System type	No 3D scene defined, no shadings			
Collector Plane Orientation		Tilt	28°	Azimuth	5°	
Models used		Transposition	Perez	Diffuse	Perez, Meteonorm	
Horizon		Free Horizon				
Near Shadings		No Shadings				
User's needs :		Unlimited load (grid)				
Power factor		Cos(phi)	0.950 lagging	Phi	-18.2°	
PV Array Characteristics						
PV module		Si-mono	Model	STP550S-C72/Vmh		
Custom parameters definition		Manufacturer		Suntech		
Number of PV modules		In series	24 modules	In parallel	16 strings	
Total number of PV modules		Nb. modules	384	Unit Nom. Power	540 Wp	
Array global power		Nominal (STC)	207 kWp	At operating cond.	185 kWp (50°C)	
Array operating characteristics (50°C)		U mpp	891 V	I mpp	207 A	
Total area		Module area	992 m²	Cell area	671 m²	
Inverter		Model		SUN2000-60KTL-HV-D1-001		
Original PVsyst database		Manufacturer		Huawei Technologies		
Characteristics		Operating Voltage	600-1450 V	Unit Nom. Power	60.0 kWac	
				Max. power (=>30°C)	66.0 kWac	
Inverter pack		Nb. of inverters	4 units	Total Power	240 kWac	
				Pnom ratio	0.86	
PV Array loss factors						
Array Soiling Losses				Loss Fraction	3.0 %	
Thermal Loss factor		Uc (const)	29.0 W/m²K	Uv (wind)	0.0 W/m²K / m/s	
Wiring Ohmic Loss		Global array res.	72 mOhm	Loss Fraction	1.5 % at STC	
LID - Light Induced Degradation				Loss Fraction	1.5 %	
Module Quality Loss				Loss Fraction	-1.3 %	
Module Mismatch Losses				Loss Fraction	1.0 % at MPP	
Strings Mismatch loss				Loss Fraction	0.10 %	
Incidence effect, ASHRAE parametrization		IAM =	1 - bo (1/cos i - 1)	bo Param.	0.05	
System loss factors						
Wiring Ohmic Loss		Wires: 3x50.0 mm²	42 m	Loss Fraction	0.5 % at STC	
Unavailability of the system		1.0 days, 3 periods		Time fraction	0.3 %	

Grid-Connected System: Main results

Project : Tawheed GP 1
Simulation variant : New simulation variant

Main system parameters

PV Field Orientation

PV modules

PV Array

Inverter

Inverter pack

User's needs

System type

No 3D scene defined, no shadings

tilt

Model

Nb. of modules

Model

Nb. of units

Unlimited load (grid)

28°

azimuth 5°

Pnom 540 Wp

Pnom total **207 kWp**

Pnom 60.0 kW ac

Pnom total **240 kW ac**

Cos(Phi) 0.950 lagging

Main simulation results

System Production

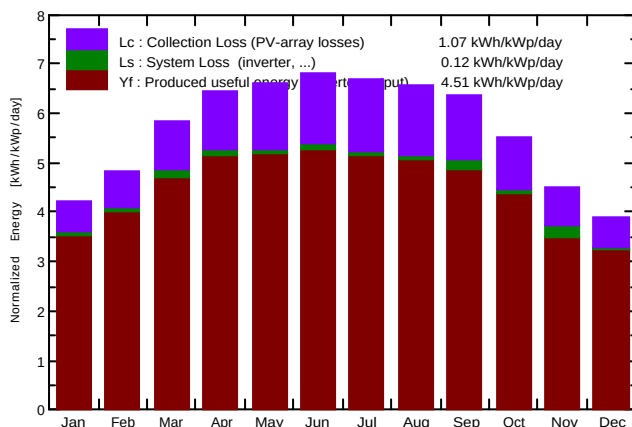
Produced Energy 341.1 MWh/year

Apparent energy 359.0 MVAh

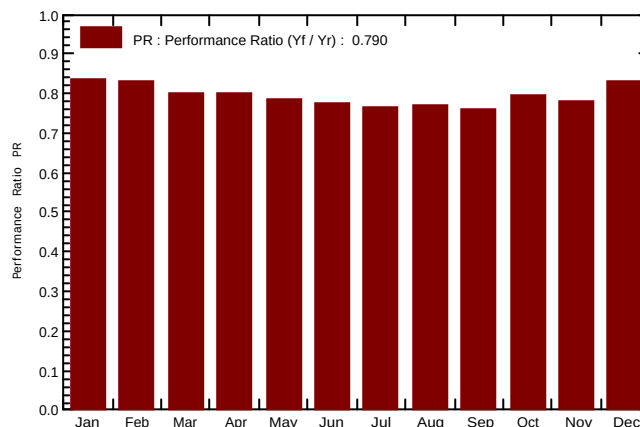
Specific prod. 1645 kWh/kWp/year

Perf. Ratio PR 79.04 %

Normalized productions (per installed kWp): Nominal power 207 kWp



Performance Ratio PR



New simulation variant Balances and main results

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray MWh	E_Grid MWh	PR
January	93.5	39.44	14.53	131.3	124.0	23.18	22.76	0.836
February	107.5	50.65	15.55	135.5	128.0	23.85	23.42	0.833
March	157.1	65.52	18.66	181.4	171.4	31.40	30.21	0.803
April	186.8	77.40	21.60	193.4	182.1	32.77	32.15	0.802
May	215.7	84.51	25.51	204.9	192.6	33.95	33.28	0.783
June	225.1	76.87	27.95	204.7	192.2	33.48	32.81	0.773
July	224.3	80.45	29.57	207.8	195.1	33.72	33.04	0.767
August	205.2	74.43	29.36	204.0	192.0	33.23	32.56	0.770
September	172.8	60.58	27.36	191.4	180.7	31.60	30.20	0.761
October	139.0	57.41	24.44	170.4	161.2	28.61	28.06	0.794
November	100.1	44.61	19.75	135.1	127.7	23.28	21.81	0.778
December	86.1	42.09	16.22	120.9	114.1	21.16	20.78	0.829
Year	1913.2	753.96	22.58	2081.0	1961.2	350.24	341.08	0.790

Legends:

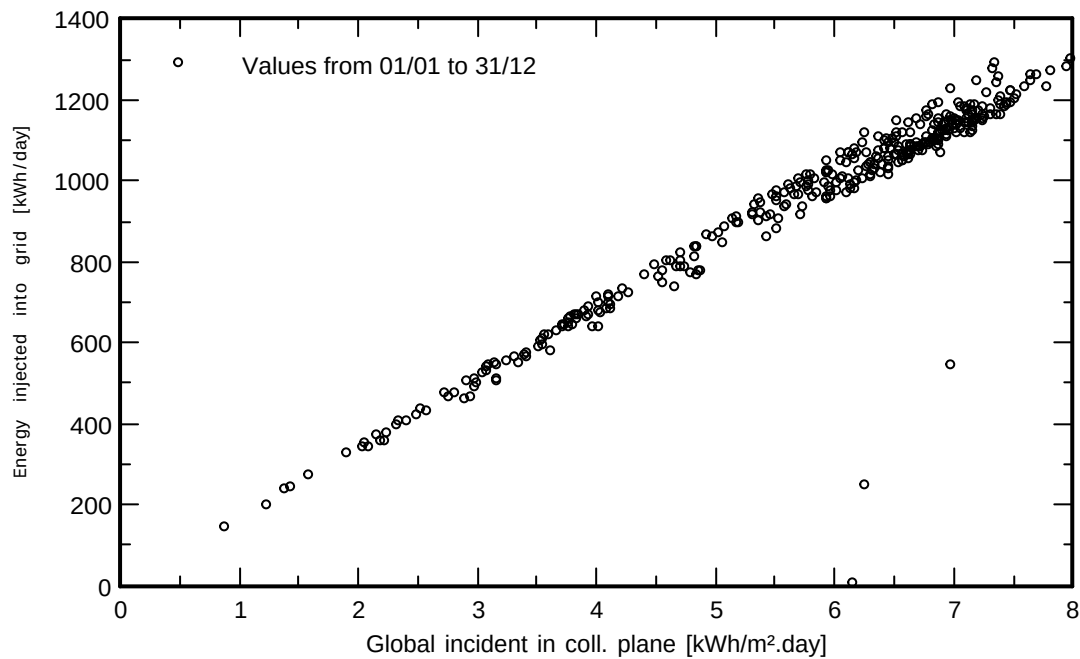
GlobHor	Horizontal global irradiation	GlobEff	Effective Global, corr. for IAM and shadings
DiffHor	Horizontal diffuse irradiation	EArray	Effective energy at the output of the array
T_Amb	Ambient Temperature	E_Grid	Energy injected into grid
GlobInc	Global incident in coll. plane	PR	Performance Ratio

Grid-Connected System: Special graphs

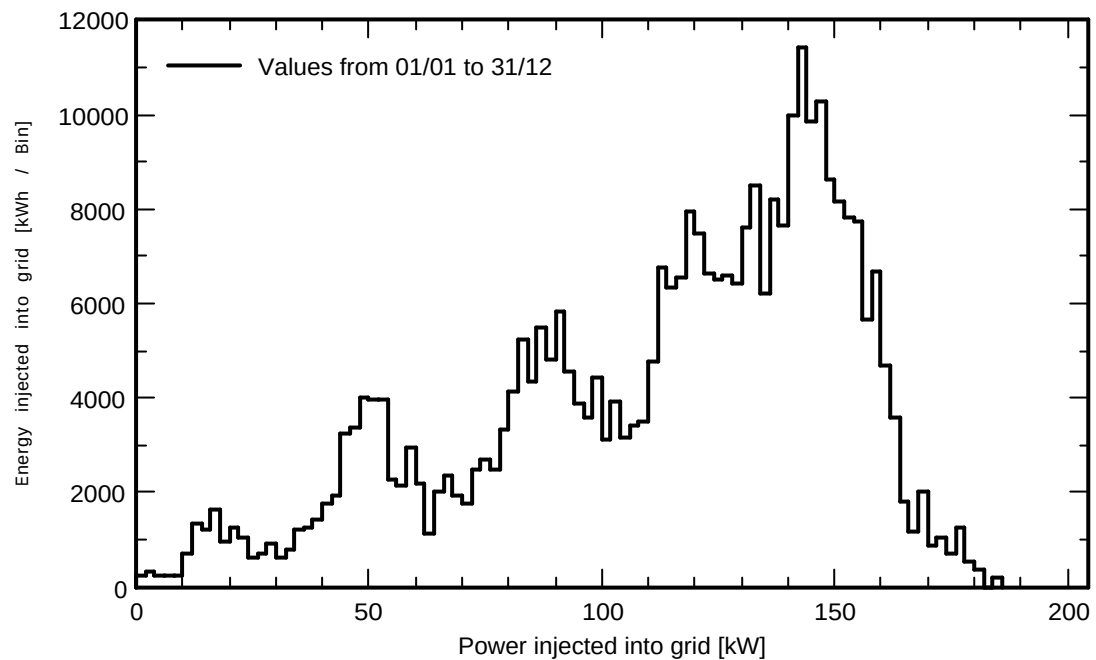
Project : Tawheed GP 1
Simulation variant : New simulation variant

Main system parameters		System type	No 3D scene defined, no shadings	
PV Field Orientation		tilt	28°	azimuth 5°
PV modules		Model	STP550S-C72/Vmh	Pnom 540 Wp
PV Array		Nb. of modules	384	Pnom total 207 kWp
Inverter		Model	SUN2000-60KTL-HV-D1-001	Pnom 60.0 kW ac
Inverter pack		Nb. of units	4.0	Pnom total 240 kW ac
User's needs		Unlimited load (grid)		Cos(Phi) 0.950 lagging

Daily Input/Output diagram



System Output Power Distribution



Grid-Connected System: Loss diagram

Project : Tawheed GP 1

Simulation variant : New simulation variant

Main system parameters

PV Field Orientation

PV modules

PV Array

Inverter

Inverter pack

User's needs

System type

tilt

No 3D scene defined, no shadings

28°

azimuth 5°

Model

STP550S-C72/Vmh

Pnom

540 Wp

Nb. of modules

384

Pnom total

207 kWp

Model

SUN2000-60KTL-HV-D1-001

Pnom

60.0 kW ac

Nb. of units

4.0

Pnom total

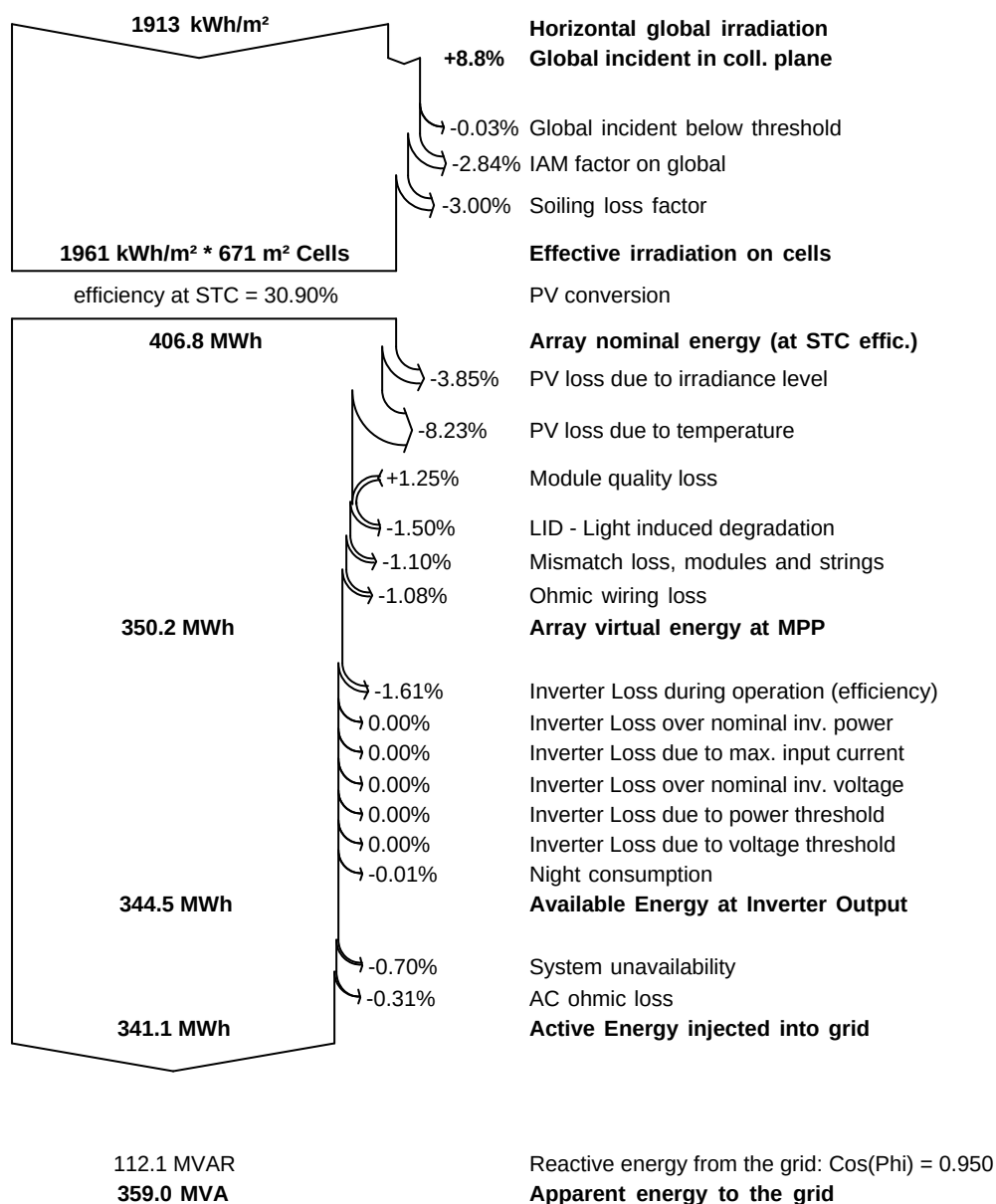
240 kW ac

Unlimited load (grid)

Cos(Phi)

0.950 lagging

Loss diagram over the whole year



Grid-Connected System: P50 - P90 evaluation

Project : Tawheed GP 1

Simulation variant : New simulation variant

Main system parameters	System type	No 3D scene defined, no shadings		
PV Field Orientation	tilt	28°	azimuth	5°
PV modules	Model	STP550S-C72/Vmh	Pnom	540 Wp
PV Array	Nb. of modules	384	Pnom total	207 kWp
Inverter	Model	SUN2000-60KTL-HV-D1-001	Pnom	60.0 kW ac
Inverter pack	Nb. of units	4.0	Pnom total	240 kW ac
User's needs	Unlimited load (grid)		Cos(Phi)	0.950 lagging

Evaluation of the Production probability forecast

The probability distribution of the system production forecast for different years is mainly dependent on the meteo data used for the simulation, and depends on the following choices:

Meteo data source	Meteonorm 7.2 (1981-2009)		
Meteo data	Kind	Not defined	Year 1995
Specified Deviation	Year deviation from aver.	3 %	
Year-to-year variability	Variance	0.5 %	

The probability distribution variance is also depending on some system parameters uncertainties

Specified Deviation	PV module modelling/parameters	1.0 %	
	Inverter efficiency uncertainty	0.5 %	
	Soiling and mismatch uncertainties	1.0 %	
	Degradation uncertainty	1.0 %	
Global variability (meteo + system)	Variance	1.9 %	(quadratic sum)

Annual production probability	Variability	6.4 MWh
	P50	341.1 MWh
	P90	332.9 MWh
	P95	330.6 MWh

Probability distribution

